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Roll No. _____

REF NO. 04121

**B. Sc. (Hons) Semester V
Supplementary Examination
2 0 2 2 – 2 3**

PHYSICS

Paper No. BPT-505 : Electromagnetic Theory

Time : 3 hours

Full Marks : 70

(Write your Roll No. at the top immediately on the receipt of this question paper)

The figures in the right-hand margin indicate marks.

Answer all questions.

All symbols have their usual meanings unless specified otherwise.

1. Answer any four of the following :

4×5

- (a) A uniform plane wave propagates in a lossless simple medium ($\varepsilon_r = 4$, $\mu_r = 1$, $\sigma = 0$) with an angular frequency 100 MHz. Calculate the wave number k .
- (b) Compute the electric field at the point $(1, 1, 0)$ for scalar potential

$$V = E_0 e^{-x} \sin \frac{xy}{4}.$$

- (c) A vector function $\vec{F} = \hat{a}_x(3y - c_1z) + \hat{a}_y(c_2x - 2z) - \hat{a}_z(c_3y + z)$ is given, where $\hat{a}_x$, $\hat{a}_y$ and $\hat{a}_z$ are unit vectors in x , y and z -directions respectively. Determine the constants c_1 , c_2 and c_3 if the function $\vec{F}$ is irrotational.
- (d) What is total reflection? Explain what happens if incident angle θ_i is greater than critical angle θ_c .
- (e) The electric field in free space is given by $\vec{E} = 50 \cos(10^8 t + \beta x) \hat{a}_y$ V/m. Calculate β and the time it takes to travel a distance $\lambda/2$, where λ is the wavelength.
- (f) Why does net charge reside on the surface of a conductor? Explain.

— (2) —

- 2. (a) Define bound currents in a magnetized material. Show that the volume current $\vec{J}_b$ and the surface current $\vec{K}_b$ are given by $\vec{J}_b = \vec{\nabla} \times \vec{M}$ and $\vec{K}_b = \vec{M} \times \hat{n}$ respectively, where $\vec{M}$ is the magnetic dipole moment per unit volume, and $\hat{n}$ is unit vector normal to the surface. **6**
- (b) Derive the boundary conditions for magnetostatic fields at an interface between two different magnetic media. **6**

— (3) —

3. (a) Discuss the propagation of electromagnetic wave in the conducting medium and show that the magnetic field intensity lags behind the electric field intensity by 45° . **8**
- (b) If $\vec{P} = 2 \sin(10t + x - \pi/4) \hat{a}_y$ and $\vec{Q}_s = e^{jk}(\hat{a}_x - \hat{a}_z) \sin(\pi y)$, determine the phasor form of $\vec{P}$ and the instantaneous form of $\vec{Q}_s$. **4**

— (4) —

4. Find the resulting potential inside and outside of a spherical shell of the radius R if a specified charge density $\sigma_0(\theta) = k \cos \theta$ is glued over the surface. **12**

Or

- (a) What were the problems in the Ampere's law and how did Maxwell fix those? Discuss its consequences. **5**
- (b) State the Poynting theorem. Show that the time average Poynting vector is $\frac{1}{2} \text{Re}(\vec{E}_s \times \vec{H}_s^*)$, where $*$ indicate complex conjugate of magnetic field $\vec{H}_s$. **7**

— (5) —

5. (a) Discuss the reflection and transmission of an obliquely incident electromagnetic wave by a boundary separated by two lossless dielectrics for the case when electromagnetic wave electric field is perpendicular to the plane of incidence. **10**
- (b) Also, derive the Brewster angle for the case (a). **4**

Or

- (a) Prove that for Transverse Magnetic (TM) mode in rectangular waveguide with perfectly conducting walls, filled with a lossless material, the expression for z -component of electric field is given by

$$E_{zs} = E_0 \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-\gamma z}$$

where symbols have their usual meanings. **7**

- (b) Determine the cutoff frequency, and phase constant (β) of TM_{11} mode for a rectangular waveguide for which $a = 1.5$ cm, $b = 0.8$ cm, $\sigma = 0$, $\mu = \mu_0$, $\varepsilon = 4\varepsilon_0$ and

$$H_x = 2 \sin\left(\frac{\pi x}{a}\right) \cos\left(\frac{3\pi y}{b}\right) \sin(\pi \times 10^{11} t - \beta z) \text{ A/m}$$

where all symbols have their usual meanings. **7**

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- **ExaminationPortal:** <https://bhu-exam.vercel.app/>
- **StudentResourceCenter:** <https://quashan-me.vercel.app/>
- **AcademicPortfolio:** <https://me-aryan.vercel.app>